Customer Churn Analysis Report

Olist E-Commerce Platform

Dataset: Olist Brazilian E-Commerce | Period: Jan 2017 – Oct 2018

Analysis Tool: SQL (PostgreSQL) + Power BI

Scope: 94,990 Customers | 10 SQL Queries | 4 Dashboard Pages

94,990	71%	R$11M	27,697
Total Customers	Churn Rate	Revenue Lost	Active Customers

Executive Summary

This report presents a comprehensive analysis of customer churn for Olist E-Commerce, covering 94,990 unique customers across 27 Brazilian states (Jan 2017 – Oct 2018).

Key Finding: 7 out of 10 customers (71%) never returned after their first purchase, resulting in R$11,037,239 in revenue that never came back. Despite impressive 10x growth in customer acquisition during 2017, the repeat purchase rate of just 3% reveals a critical retention gap.

This analysis identifies root causes — late delivery, low review scores, geographic disparities — and provides 6 actionable recommendations to recover R$1.1M+ in revenue.

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1. Business Overview & Dataset Scope

Olist is Brazil's largest e-commerce marketplace connecting small and medium sellers to major retail platforms. This analysis uses the publicly available Olist dataset covering orders from January 2017 to October 2018.

Parameter	Details
Dataset Period	January 2017 – October 2018
Total Unique Customers	94,990
Total Orders	~99,000+
Geographic Coverage	27 Brazilian States
Analysis Tool	PostgreSQL (SQL) + Power BI
SQL Queries Built	10 analytical queries + 10 SQL views
Churn Definition	Customer with no order in last 180 days (from Oct 17, 2018)
Total Revenue — Active Customers	R$4,701,898
Total Revenue — Churned Customers	R$11,037,239

2. Key Performance Metrics — At a Glance

94,990	**71%**	**R$11.0M**	**27,697**
Total Customers	Churn Rate	Revenue Lost	Active Customers
All Time	67,293 customers	70% of total revenue	29% of total

3.0%	**R$308**	**8.3%**	**4.1/5**
Repeat Purchase Rate	Repeat Avg LTV	Late Delivery Rate	Avg Review Score
2,888 customers	vs R$161 one-time	7,826 orders late	98,410 reviews

The data reveals a fundamental business problem: while Olist grew its customer base 10x from January to November 2017, **7 out of 10 customers never returned**. The platform spent resources acquiring new customers while losing existing ones — resulting in R$11M in one-time revenue that will never recur without intervention.

3. SQL Query 1: Overall Churn Rate

Business Question: How many customers stopped buying from Olist?

Churn is defined as a customer with no order in the last 180 days (reference: October 17, 2018).

SQL Query:

```
CREATE OR REPLACE VIEW olist.v_churn_rate AS
SELECT churn_status, COUNT(*) AS total_customers,
ROUND(COUNT(*)::NUMERIC / 96096 * 100, 1) AS percentage
FROM (
SELECT c.customer_unique_id,
CASE
WHEN DATE '2018-10-17' - MAX(o.order_purchase_timestamp)::DATE > 180
THEN 'Churned' ELSE 'Active'
END AS churn_status
FROM olist.customers c
JOIN olist.orders o ON c.customer_id = o.customer_id
WHERE o.order_status NOT IN ('canceled', 'unavailable')
GROUP BY c.customer_unique_id) AS churn_summary
GROUP BY churn_status;
```

Query Output:

churn_status	total_customers	percentage
Churned	67,293	70.0%
Active	27,697	28.8%

Business Impact

→ 7 out of 10 customers never came back after their first order.

→ 70% churn rate across 94,990 customers — Olist has no retention strategy in place.

→ Fixing this directly increases revenue without spending more on acquiring new customers.

4. SQL Query 2: Churn Segmentation

Business Question: Who are the churned customers and how long ago did they stop buying?

4-tier segmentation: Active (<=90 days), At Risk (90–180), Churned Recent (180–365), Churned Long Term (365+).

SQL Query:

```
CREATE OR REPLACE VIEW olist.v_customer_churn_status AS
SELECT c.customer_unique_id, c.customer_state,
COUNT(DISTINCT o.order_id) AS total_orders,
ROUND(SUM(p.payment_value)::NUMERIC, 2) AS total_spend,
MAX(o.order_purchase_timestamp)::DATE AS last_order_date,
DATE '2018-10-17' - MAX(o.order_purchase_timestamp)::DATE AS recency_days,
CASE
WHEN DATE '2018-10-17' - MAX(o.order_purchase_timestamp)::DATE <= 90 THEN 'Active'
WHEN DATE '2018-10-17' - MAX(o.order_purchase_timestamp)::DATE <= 180 THEN 'At Risk'
WHEN DATE '2018-10-17' - MAX(o.order_purchase_timestamp)::DATE <= 365 THEN 'Churned - Recent'
ELSE 'Churned - Long Term'
END AS churn_status
FROM olist.customers c
JOIN olist.orders o ON c.customer_id = o.customer_id
JOIN olist.order_payments p ON o.order_id = p.order_id
WHERE o.order_status NOT IN ('canceled', 'unavailable')
GROUP BY c.customer_unique_id, c.customer_state;
```

Query Output:

churn_status	total_customers	avg_spend (R$)	avg_days_since_last_order
Churned - Recent	39,359	162.35	269
Churned - Long Term	27,966	166.27	483
At Risk	18,238	172.39	138
Active	9,464	164.33	72

Business Impact

→ 39,359 customers churned in the last 6–12 months (Churned Recent).

→ 27,966 customers have been gone for over a year (Churned Long Term).

→ At Risk segment of 18,238 customers is the most valuable — still reachable with a discount or reminder.

→ Average spend is similar across all groups (~R$162–172) — all segments are worth recovering.

→ Churned Long Term customers need a strong win-back campaign: special offer or re-engagement email.

5. SQL Query 3 & 4: Repeat Purchase Rate & Customer LTV

Business Question: How many customers came back for a second order? How much more do they spend?

Repeat customer = more than 1 distinct order. LTV = total sum of all payment values per customer.

SQL Query:

```
-- Query 3: Repeat Purchase Rate
SELECT customer_type, COUNT(*) AS total_customers,
ROUND(COUNT(*)::NUMERIC / 96096 * 100, 1) AS percentage,
ROUND(AVG(total_spend)::NUMERIC, 2) AS avg_spend
FROM (SELECT c.customer_unique_id, SUM(p.payment_value) AS total_spend,
CASE WHEN COUNT(DISTINCT o.order_id) > 1
THEN 'Repeat Customer' ELSE 'One-Time Customer' END AS customer_type
FROM olist.customers c
JOIN olist.orders o ON c.customer_id = o.customer_id
JOIN olist.order_payments p ON o.order_id = p.order_id
WHERE o.order_status NOT IN ('canceled', 'unavailable')
GROUP BY c.customer_unique_id) AS summary
GROUP BY customer_type;

-- Query 4: Customer Value Comparison (avg_orders, avg_lifetime_value)
```

Query Output:

customer_type	total_customers	percentage	avg_LTV (R$)	avg_orders
One-Time Customer	92,101	95.8%	161.22	1.0
Repeat Customer	2,888	3.0%	308.36	2.1

Business Impact

→ Only 3 out of 100 customers placed a second order (3.0%).

→ A repeat customer spends 1.9x more than a one-time customer (R$308 vs R$161).

→ Repeat customers place 2.1 orders on average vs 1.0 for one-time buyers.

→ Investing in retention is more profitable than spending on acquiring new customers.

6. SQL Query 5: Churn Rate by State

Business Question: Which states have the highest customer churn rate?

Churn rate calculated per state = churned customers / total customers * 100.

SQL Query:

```
CREATE OR REPLACE VIEW olist.v_churn_by_state AS
SELECT customer_state, COUNT(*) AS total_customers,
SUM(CASE WHEN is_churned = 1 THEN 1 ELSE 0 END) AS churned_customers,
SUM(CASE WHEN is_churned = 0 THEN 1 ELSE 0 END) AS active_customers,
ROUND(SUM(CASE WHEN is_churned = 1 THEN 1 ELSE 0 END)::NUMERIC
/ COUNT(*) * 100, 1) AS churn_rate_pct
FROM (SELECT c.customer_unique_id, c.customer_state,
CASE WHEN DATE '2018-10-17' - MAX(o.order_purchase_timestamp)::DATE > 180
THEN 1 ELSE 0 END AS is_churned
FROM olist.customers c JOIN olist.orders o ON c.customer_id = o.customer_id
WHERE o.order_status NOT IN ('canceled', 'unavailable')
GROUP BY c.customer_unique_id, c.customer_state) AS customer_churn
GROUP BY customer_state ORDER BY churn_rate_pct DESC;
```

Query Output:

State	Total Customers	Churned	Active	Churn Rate (%)
AC	77	63	14	81.8%
AL	399	310	89	77.7%
PA	944	729	215	77.2%
MA	715	549	166	76.8%
RO	234	178	56	76.1%
AP	67	51	16	76.1%
CE	1,301	981	320	75.4%
RR	44	33	11	75.0%

Business Impact

→ AC state has the highest churn at 81.8%, followed by AL (77.7%) and PA (77.2%).

→ Northern and remote states dominate the top churn list.

→ Root cause: longer delivery times and higher freight costs in these regions.

→ SP has the most customers but lower churn rate — better logistics = better retention.

→ Adding regional sellers or logistics hubs in high-churn states can reduce churn significantly.

7. SQL Query 6: Revenue Lost to Churn

Business Question: How much revenue was lost because customers did not come back?

Revenue is split between churned (180+ days inactive) and active customers. Recovery potential = 10% of churned revenue.

SQL Query:

```
CREATE OR REPLACE VIEW olist.v_revenue_churn AS
SELECT customer_status, COUNT(DISTINCT customer_unique_id) AS total_customers,
ROUND(SUM(total_spend)::NUMERIC, 2) AS total_revenue,
ROUND(AVG(total_spend)::NUMERIC, 2) AS avg_order_value
FROM (SELECT c.customer_unique_id, SUM(p.payment_value) AS total_spend,
CASE WHEN DATE '2018-10-17' - MAX(o.order_purchase_timestamp)::DATE > 180
THEN 'Churned' ELSE 'Active' END AS customer_status
FROM olist.customers c
JOIN olist.orders o ON c.customer_id = o.customer_id
JOIN olist.order_payments p ON o.order_id = p.order_id
WHERE o.order_status NOT IN ('canceled', 'unavailable')
GROUP BY c.customer_unique_id) AS customer_summary
GROUP BY customer_status;
```

Query Output:

customer_status	total_customers	total_revenue (R$)	avg_order_value (R$)
Active	27,697	4,701,898.14	169.76
Churned	67,292	11,037,238.87	164.02

Business Impact

→ Churned customers generated R$11,037,239 but never returned — 70% of total revenue.

→ Active customers generated only R$4,701,898 (30% of total revenue).

→ If 10% of churned customers were retained = R$1,103,724 in additional revenue with zero acquisition cost.

8. SQL Query 7: Review Score vs Churn Rate

Business Question: Do customers who gave a low rating churn more than those who gave a high rating?

Churn rate calculated for each review score group (1–5 stars). A customer may appear in multiple groups.

SQL Query:

```
CREATE OR REPLACE VIEW olist.v_review_vs_churn AS
SELECT review_score, COUNT(DISTINCT customer_unique_id) AS total_customers,
SUM(CASE WHEN is_churned = 1 THEN 1 ELSE 0 END) AS churned_customers,
ROUND(SUM(CASE WHEN is_churned = 1 THEN 1 ELSE 0 END)::NUMERIC
/ COUNT(DISTINCT customer_unique_id) * 100, 1) AS churn_rate_pct
FROM (SELECT c.customer_unique_id, r.review_score,
CASE WHEN DATE '2018-10-17' - MAX(o.order_purchase_timestamp)::DATE > 180
THEN 1 ELSE 0 END AS is_churned
FROM olist.customers c
JOIN olist.orders o ON c.customer_id = o.customer_id
JOIN olist.order_reviews r ON o.order_id = r.order_id
WHERE o.order_status NOT IN ('canceled', 'unavailable')
AND r.review_score > 0
GROUP BY c.customer_unique_id, r.review_score) AS review_churn
GROUP BY review_score ORDER BY review_score;
```

Query Output:

Review Score	Total Customers	Churned Customers	Churn Rate (%)
1 Star	10,338	8,091	78.3%
2 Stars	3,020	2,290	75.8%
3 Stars	7,967	5,963	74.8%
4 Stars	18,748	13,471	71.9%
5 Stars	55,316	37,920	68.6%

Business Impact

→ 1-star reviewers have a 78.3% churn rate — highest of any group.

→ Even 5-star reviewers churn at 68.6%, showing a platform-wide retention problem.

→ Every 1-point drop in review score increases churn by approximately 3%.

→ A 1-star review is an early warning sign of churn — fast resolution can prevent permanent loss.

9. SQL Query 8: Late Delivery Impact on Churn

Business Question: Do customers who received a late delivery churn more than on-time customers?

Late delivery = actual delivery date exceeded estimated delivery date. Churn rate and avg review score compared across groups.

SQL Query:

```
CREATE OR REPLACE VIEW olist.v_delivery_vs_churn AS
SELECT delivery_status, COUNT(DISTINCT customer_unique_id) AS total_customers,
SUM(CASE WHEN is_churned = 1 THEN 1 ELSE 0 END) AS churned_customers,
ROUND(SUM(CASE WHEN is_churned = 1 THEN 1 ELSE 0 END)::NUMERIC
/ COUNT(DISTINCT customer_unique_id) * 100, 1) AS churn_rate_pct,
ROUND(AVG(avg_review_score)::NUMERIC, 2) AS avg_review_score
FROM (SELECT c.customer_unique_id,
CASE WHEN o.is_late_delivery = TRUE
THEN 'Late Delivery' ELSE 'On Time Delivery' END AS delivery_status,
CASE WHEN DATE '2018-10-17' - MAX(o.order_purchase_timestamp)::DATE > 180
THEN 1 ELSE 0 END AS is_churned,
AVG(r.review_score) AS avg_review_score
FROM olist.customers c
JOIN olist.orders o ON c.customer_id = o.customer_id
LEFT JOIN olist.order_reviews r ON o.order_id = r.order_id
WHERE o.is_valid_delivered = TRUE
GROUP BY c.customer_unique_id, o.is_late_delivery) AS delivery_churn
GROUP BY delivery_status;
```

Query Output:

delivery_status	total_customers	churned	churn_rate_pct	avg_review_score
Late Delivery	7,771	6,103	78.5%	2.57
On Time Delivery	85,891	60,140	70.0%	4.29

Business Impact

→ Late delivery customers churn at 78.5% vs 70.0% for on-time — an 8.5pp difference.

→ Late delivery drives lower review scores: avg 2.57 vs 4.29 for on-time customers.

→ 8.3% of deliveries are late but cause disproportionate damage to retention and reviews.

→ Improving delivery speed is one of the fastest ways to reduce churn.

10. SQL Query 9: Monthly New Customers Trend

Business Question: How many new customers joined Olist every month? Is the business growing?

New customer = unique customer_unique_id whose first-ever order falls in that calendar month.

SQL Query:

```
CREATE OR REPLACE VIEW olist.v_monthly_new_customers AS
SELECT DATE_TRUNC('month', first_order_date)::DATE AS order_month,
COUNT(*) AS new_customers
FROM (SELECT c.customer_unique_id,
MIN(o.order_purchase_timestamp)::DATE AS first_order_date
FROM olist.customers c JOIN olist.orders o ON c.customer_id = o.customer_id
WHERE o.order_status NOT IN ('canceled', 'unavailable')
GROUP BY c.customer_unique_id) AS first_orders
WHERE first_order_date >= '2017-01-01'
GROUP BY order_month ORDER BY order_month;
```

Query Output:

order_month	new_customers
Jan 2017	752
Feb 2017	1,690
May 2017	3,541
Aug 2017	4,130
Nov 2017	7,190 — Black Friday Peak
Jan 2018	6,951
Apr 2018	6,698
Aug 2018	6,042

Business Impact

→ New customer acquisition grew ~9.6x from Jan 2017 (752) to Nov 2017 (7,190).

→ November 2017 Black Friday peak = 7,190 new customers in a single month.

→ Growth stabilized at ~6,000–7,000 new customers per month through 2018.

→ High acquisition + 3% repeat rate = business keeps spending to replace lost customers.

→ Retention must grow alongside acquisition for sustainable business growth.

11. SQL Query 10: Top Categories for Repeat Customers

Business Question: What do loyal customers buy most? Which product categories drive repeat purchases?

Only customers with more than 1 distinct order included. Categories ranked by total order count.

SQL Query:

```
CREATE OR REPLACE VIEW olist.v_repeat_customer_categories AS
SELECT COALESCE(p.product_category_english, 'unknown') AS category,
COUNT(DISTINCT o.order_id) AS total_orders,
COUNT(DISTINCT c.customer_unique_id) AS total_customers,
ROUND(SUM(oi.price)::NUMERIC, 2) AS total_revenue,
ROUND(AVG(oi.price)::NUMERIC, 2) AS avg_price
FROM olist.customers c
JOIN olist.orders o ON c.customer_id = o.customer_id
JOIN olist.order_items oi ON o.order_id = oi.order_id
JOIN olist.products p ON oi.product_id = p.product_id
WHERE o.order_status NOT IN ('canceled', 'unavailable')
AND c.customer_unique_id IN (
SELECT c2.customer_unique_id FROM olist.customers c2
JOIN olist.orders o2 ON c2.customer_id = o2.customer_id
WHERE o2.order_status NOT IN ('canceled', 'unavailable')
GROUP BY c2.customer_unique_id HAVING COUNT(DISTINCT o2.order_id) > 1)
GROUP BY category ORDER BY total_orders DESC LIMIT 10;
```

Query Output:

Category	Orders	Customers	Revenue (R$)	Avg Price (R$)
bed_bath_table	867	595	97,874	88.57
sports_leisure	603	401	74,863	111.07
furniture_decor	586	455	63,219	78.05
health_beauty	506	350	55,601	100.54
computers_accessories	411	279	62,496	114.46
housewares	348	285	34,704	82.04
watches_gifts	278	202	52,569	165.83
fashion_bags_accessories	227	161	16,830	64.73

Business Impact

→ bed_bath_table is the top repeat category — 867 orders from 595 unique repeat customers.

→ Top 4 categories are all home & lifestyle — this is where customer loyalty is built.

→ Personalized recommendations to one-time buyers in these categories within 45 days drives conversion.

→ A 5% improvement in repeat purchase rate = estimated R$800K in additional revenue.

12. Power BI Dashboard Screenshots

The following screenshots show the interactive Power BI dashboard built from the SQL analysis. The dashboard has 4 pages: Overview, Churn Analysis, Retention Analysis, and Delivery & Reviews.

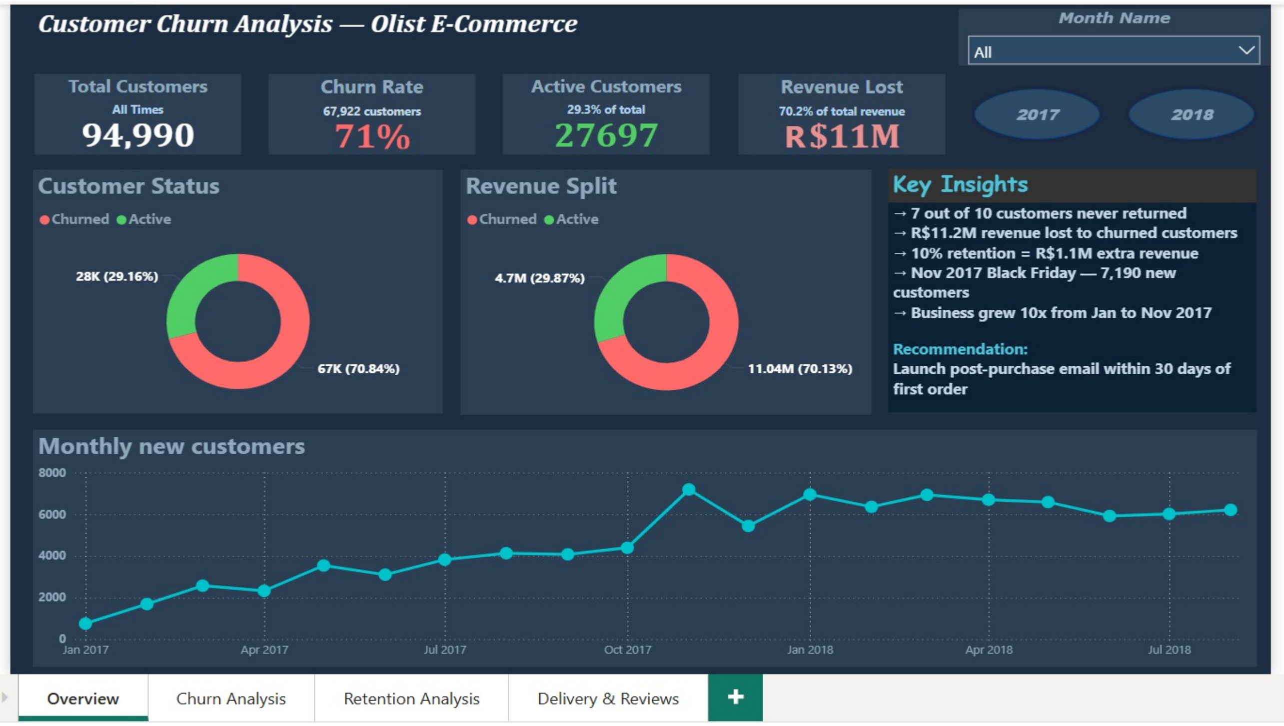

Figure 1: Overview Page — Total Customers, Churn Rate, Revenue Lost & Monthly New Customer Trend

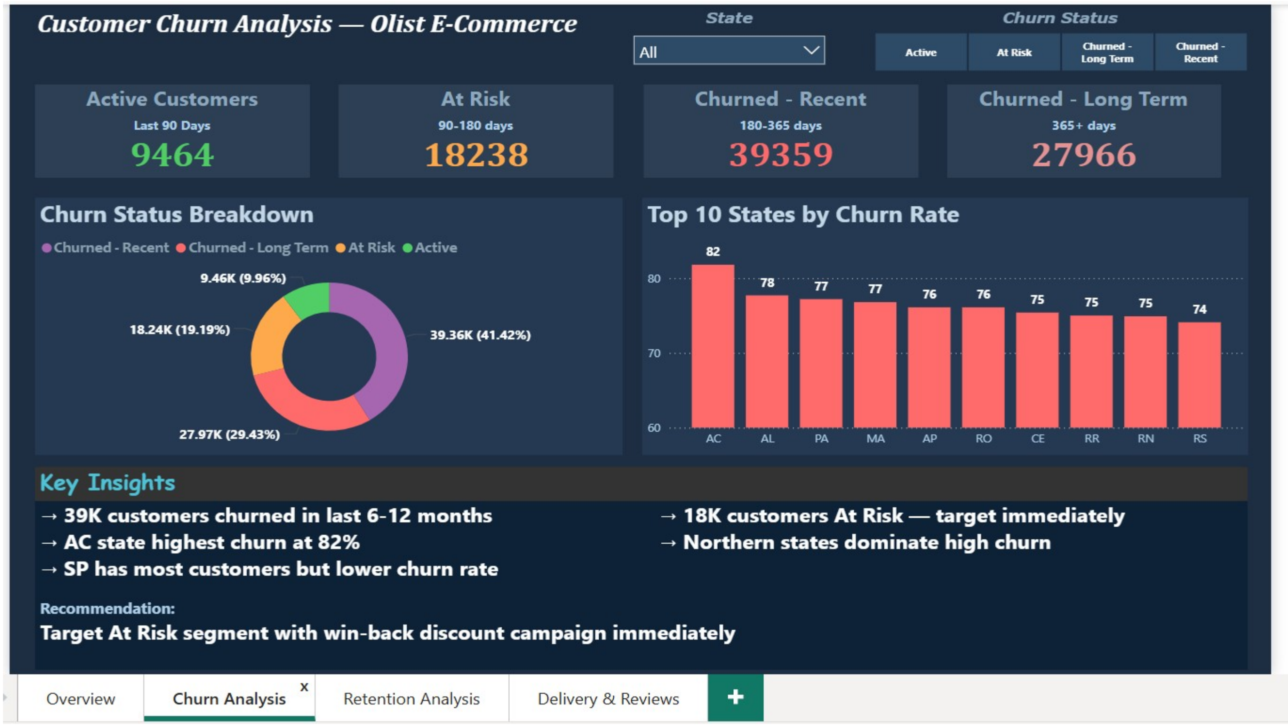

Figure 2: Churn Analysis Page — Segment Breakdown by Status & Top 10 States by Churn Rate

Figure 3: Retention Analysis Page — Repeat Purchase Rate, LTV Comparison & Top Repeat Categories

Figure 4: Delivery & Reviews Page — Late Delivery vs Churn & Review Score vs Churn Rate

13. Recommendations & Next Steps

Based on the SQL analysis above, the following 6 recommendations are prioritized by revenue impact and ease of implementation.

01 Post-Purchase Email Campaign (Within 30 Days)

Priority: HIGH — Implement Immediately

R$1.1M+ potential

Send a targeted follow-up email to every new customer within 30 days of their first order with personalized product recommendations from top repeat categories: bed_bath_table, sports_leisure, and health_beauty. This directly targets the 97% one-time buyers who showed purchase intent but never returned.

02 Win-Back Campaign for At-Risk Customers

Priority: HIGH — Implement Immediately

18,238 customers to target

Target the 18,238 At-Risk customers (last order 90–180 days ago) with a discount voucher or personalized reminder. These customers have not yet fully churned and are the easiest and cheapest to recover. Avg spend of R$172 makes them high-value targets.

03 Penalize Sellers with High Late Delivery Rate

Priority: MEDIUM — Within 30 Days

Closes 8.5pp churn gap

Sellers with late delivery rate above 10% should face reduced search ranking, fee penalties, or mandatory performance improvement plans. Late deliveries cause 78.5% churn vs 70.0% on-time — an 8.5pp difference. They also drag avg review score from 4.29 to 2.57.

04 1-Star Review Alert & Recovery System

Priority: MEDIUM — Within 30 Days

78.3% of 1-star reviewers churn

Implement an automated alert that flags 1-star reviews in real time and triggers a customer service follow-up. 1-star reviewers churn at 78.3% — highest of any group. A fast resolution (refund, replacement, or apology voucher) before they permanently disengage can save a significant portion of these customers.

05 Improve Logistics for Northern States

Priority: MEDIUM — Strategic Planning

AC: 82% churn, AL: 78%

States AC, AL, PA, MA, RO, AP all show 76–82% churn — well above national average. The primary drivers are longer delivery times and higher freight costs in remote regions. Adding regional sellers, fulfilment hubs, or logistics partnerships in these states will reduce churn and improve review scores.

06 Loyalty Program / Repeat Purchase Incentive

Priority: STRATEGIC — 60–90 Days

5% improvement = R$800K extra

Introduce a points-based loyalty system or cashback for second orders. With only 3% repeat rate, even a 5% improvement generates ~R$800K in additional revenue. Repeat customers spend 1.9x more on average (R$308 vs R$161), making loyalty investment highly profitable.

Key Takeaway

Olist grew 10x in acquisition but failed to retain customers. 7 out of 10 never came back. The fix is not more acquisition — it is retention. A post-purchase email + win-back campaign alone can unlock **R$1.1M in recovery revenue with zero new customer acquisition cost.**